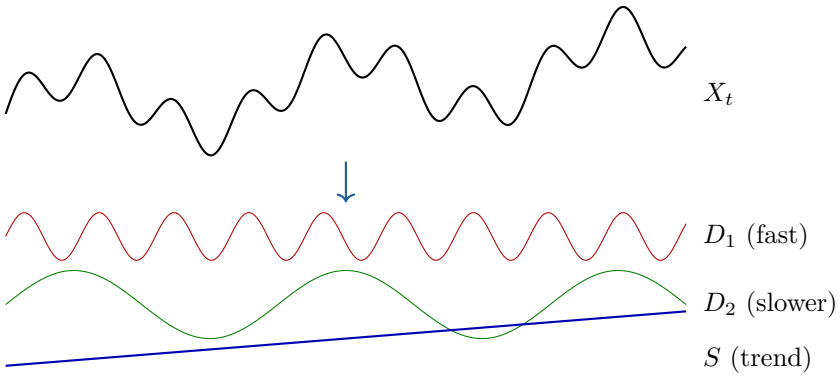
The Maximal Overlap Discrete Wavelet Transform

An in-depth explanation with worked examples

Theory, hand calculations, properties and its role in WaveCatBoost



Companion notes for the paper
A Novel Hybrid Approach for Efficiently Forecasting Air Quality Data (WaveCatBoost),
IEEE Sensors Letters, 2025

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1 Why decompose a time series at all?

An hourly air-quality series such as $PM_{2.5}$ is a superposition of behaviours that live on very different time scales:

- **Very fast variation**: sensor noise and sudden spikes (a passing truck, a fire), lasting an hour or two.
- **Intra-day structure**: morning and evening traffic peaks, the 12-hour and 24-hour cycles driven by traffic and boundary-layer meteorology.
- **Multi-day structure**: weather systems, weekday/weekend differences.
- **Slow trend**: seasonal build-up of pollution, long-term drift.

A single forecasting model fed with the raw series must learn all of these at once, even though they follow very different dynamics. The idea behind wavelet-based hybrid forecasting is:

The divide-and-conquer principle

Split the series into components that each contain *one* time scale, forecast each component with its own model, then add the component forecasts back together.

$$X_t \xrightarrow{\text{MODWT}} \{D_{1,t}, \ldots, D_{J,t}, S_{J,t}\} \xrightarrow{\text{forecast each}} \{\hat{D}_1, \ldots, \hat{D}_J, \hat{S}_J\} \xrightarrow{\text{combine}} \hat{X}.$$

1.1 Why not just use the Fourier transform?

The Fourier transform writes a signal as a sum of sines and cosines that extend over the *entire* record. It tells you *which* frequencies are present but not *when* they occur. A pollution spike at hour 150 is spread across all Fourier coefficients, and a daily cycle whose amplitude changes from summer to winter is described only by its average amplitude. Air-quality series are **non-stationary**: their mean, variance and periodic structure change over time. We need a tool that is localised in *both* time and scale. That tool is the wavelet transform.

2 Wavelet fundamentals

2.1 What is a wavelet?

A wavelet $\psi(t)$ is a "small wave": a function that oscillates but dies out quickly. It satisfies

$$\int_{-\infty}^{\infty} \psi(t)\,dt = 0 \qquad \text{(it has zero mean, so it measures } \textit{change}\text{)}, \qquad \int_{-\infty}^{\infty} \psi^2(t)\,dt = 1.$$

By *stretching* the wavelet we look at slow changes, by *squeezing* it we look at fast changes, and by *sliding* it along the time axis we locate those changes in time.

The simplest wavelet, and the one used in WaveCatBoost, is the **Haar wavelet**:

$$\psi^{\text{Haar}}(t) = \begin{cases} +1, & 0 \le t < \frac{1}{2}, \\ -1, & \frac{1}{2} \le t < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Correlating a signal with a Haar wavelet simply computes "(average of the recent half) minus (average of the older half)". It is a *change detector*.

2.2 From functions to filters

For discrete data (one value per hour) we never use $\psi(t)$ directly. Instead we use two short digital filters:

- the **wavelet filter** $\{h_l\}_{l=0}^{L-1}$: a *high-pass* filter that extracts detail (change);
- the **scaling filter** $\{g_l\}_{l=0}^{L-1}$: a *low-pass* filter that extracts the smooth part (local average).

A valid (orthonormal) wavelet filter must satisfy

$$\sum_l h_l = 0 \qquad \text{(zero mean: a constant signal has no detail)} \tag{1}$$

$$\sum_l h_l^2 = 1 \qquad \text{(unit energy)} \tag{2}$$

$$\sum_l h_l\, h_{l+2n} = 0 \quad (n \neq 0) \qquad \text{(orthogonal to its even shifts)} \tag{3}$$

and the scaling filter is obtained from it by the *quadrature mirror* relation $g_l = (-1)^{l+1} h_{L-1-l}$. Condition (3) is the orthogonality relation that appears in the WaveCatBoost paper (written there as $\sum_m p_m p_{m+2n} = 0$ with p the wavelet filter and q the scaling filter).

Example: the Haar filters

For Haar, $L = 2$ and

$$h = \left(\tfrac{1}{\sqrt{2}}, -\tfrac{1}{\sqrt{2}}\right), \qquad g = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}\right).$$

Check: $\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$ ✓, $\frac{1}{2} + \frac{1}{2} = 1$ ✓. With $L = 2$ there are no nonzero even shifts that overlap, so (3) holds trivially. Applied to a pair (X_{t-1}, X_t):

wavelet: $\frac{1}{\sqrt{2}}(X_t - X_{t-1})$ (a difference), scaling: $\frac{1}{\sqrt{2}}(X_t + X_{t-1})$ (a scaled average).

3 The ordinary DWT and its problems

The classical **discrete wavelet transform** (DWT) uses Mallat's *pyramid algorithm*: filter the series with h and g, then **keep only every second value** (downsample by 2). The smooth half is fed into the next level and the process repeats. For Haar, level 1 is

$$W_{1,k}^{\mathrm{DWT}} = \frac{X_{2k+1} - X_{2k}}{\sqrt{2}}, \qquad V_{1,k}^{\mathrm{DWT}} = \frac{X_{2k+1} + X_{2k}}{\sqrt{2}}, \qquad k = 0, \ldots, \tfrac{N}{2} - 1.$$

Each level halves the number of coefficients, so N values become $N/2 + N/4 + \ldots$ coefficients, N in total. That is efficient for compression, but it causes three problems for time-series forecasting:

1. **Length restriction.** A level-J DWT needs N to be a multiple of 2^J.
2. **Shift variance.** Because pairs are formed as $(X_0, X_1), (X_2, X_3), \ldots$, starting the series one hour later changes the pairing and therefore changes the coefficients *completely*, not just by a shift (see Section 8.2).
3. **Misalignment with time.** The level-j coefficients are spaced 2^j hours apart, so they do not line up with the original hourly time axis. You cannot build lag features such as $D_{j,t-1}, D_{j,t-2}$ that are synchronised with X_t.

The MODWT removes all three problems by *not downsampling*.

4 The MODWT: definition

4.1 Rescaled filters

Because nothing is thrown away, the filters must be rescaled to keep energy balanced. The MODWT filters are

$$\boxed{\tilde{h}_l = \frac{h_l}{\sqrt{2}}, \qquad \tilde{g}_l = \frac{g_l}{\sqrt{2}}} \qquad \text{so for Haar:} \qquad \tilde{h} = (\tfrac{1}{2}, -\tfrac{1}{2}), \quad \tilde{g} = (\tfrac{1}{2}, \tfrac{1}{2}).$$

4.2 The MODWT pyramid algorithm

Start with $V_{0,t} = X_t$. For each level $j = 1, 2, \ldots, J$ compute, for *every* $t = 0, \ldots, N-1$,

$$\boxed{W_{j,t} = \sum_{l=0}^{L-1} \tilde{h}_l \, V_{j-1,\,(t-2^{j-1}l) \bmod N}, \qquad V_{j,t} = \sum_{l=0}^{L-1} \tilde{g}_l \, V_{j-1,\,(t-2^{j-1}l) \bmod N}.} \tag{4}$$

Three things are worth noticing:

- **No downsampling**: every level has exactly N coefficients, one per time point.
- **Growing gaps**: at level j the filter taps are 2^{j-1} steps apart (1, 2, 4, 8, ...). This is the *à trous* ("with holes") trick: it is equivalent to inserting $2^{j-1} - 1$ zeros between the filter coefficients. That is how deeper levels see longer time spans.
- **Circular boundary**: indices are taken mod N, so the series is treated as if it wrapped around: X_{-1} means X_{N-1}. This affects the first few coefficients of each level (Section 8.5).

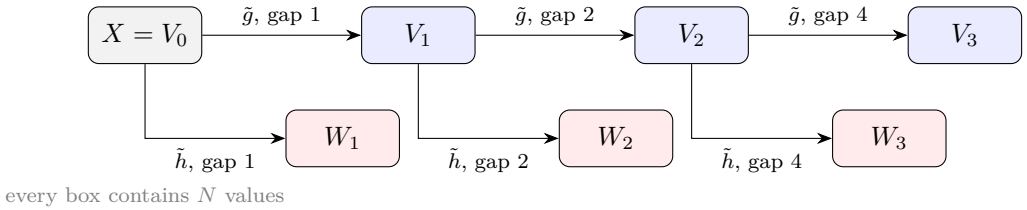


Figure 1: The MODWT pyramid. At each level the current smooth series is split into a detail series W_j and a smoother series V_j. The smooth series is split again, with the filter gap doubling each time.

4.3 The Haar MODWT in plain words

Plugging the Haar filters into (4) gives very simple recursions:

$$\boxed{W_{j,t} = \frac{V_{j-1,t} - V_{j-1,t-2^{j-1}}}{2}, \qquad V_{j,t} = \frac{V_{j-1,t} + V_{j-1,t-2^{j-1}}}{2}.} \tag{5}$$

Unrolling the recursion gives an even more intuitive description. $V_{j,t}$ is the average of the last 2^j observations, and $W_{j,t}$ is half the difference of two adjacent block averages:

$$V_{j,t} = \frac{1}{2^j} \sum_{l=0}^{2^j-1} X_{t-l}, \tag{6}$$

$$W_{j,t} = \frac{1}{2}\Bigg[\underbrace{\frac{1}{2^{j-1}} \sum_{l=0}^{2^{j-1}-1} X_{t-l}}_{\text{mean of most recent } 2^{j-1}} - \underbrace{\frac{1}{2^{j-1}} \sum_{l=2^{j-1}}^{2^j-1} X_{t-l}}_{\text{mean of the } 2^{j-1} \text{ before that}} \Bigg]. \tag{7}$$

Intuition for the Haar MODWT

- $V_{j,t}$ is a moving average over a window of 2^j hours.
- $W_{j,t}$ is half the difference between two adjacent block averages, each 2^{j-1} hours long. It asks: *did the level change between the previous block and the latest block?*
- W_1 compares single hours, W_2 compares 2-hour blocks, W_3 compares 4-hour blocks, and so on.

Equivalently, each level can be computed in one step with an *equivalent filter* applied directly to X: $W_{j,t} = \sum_l \tilde{h}_{j,l} X_{t-l}$ and $V_{j,t} = \sum_l \tilde{g}_{j,l} X_{t-l}$, where the equivalent filters have width $L_j = (2^j - 1)(L - 1) + 1$, which is 2^j for Haar. Figure 2 shows them.

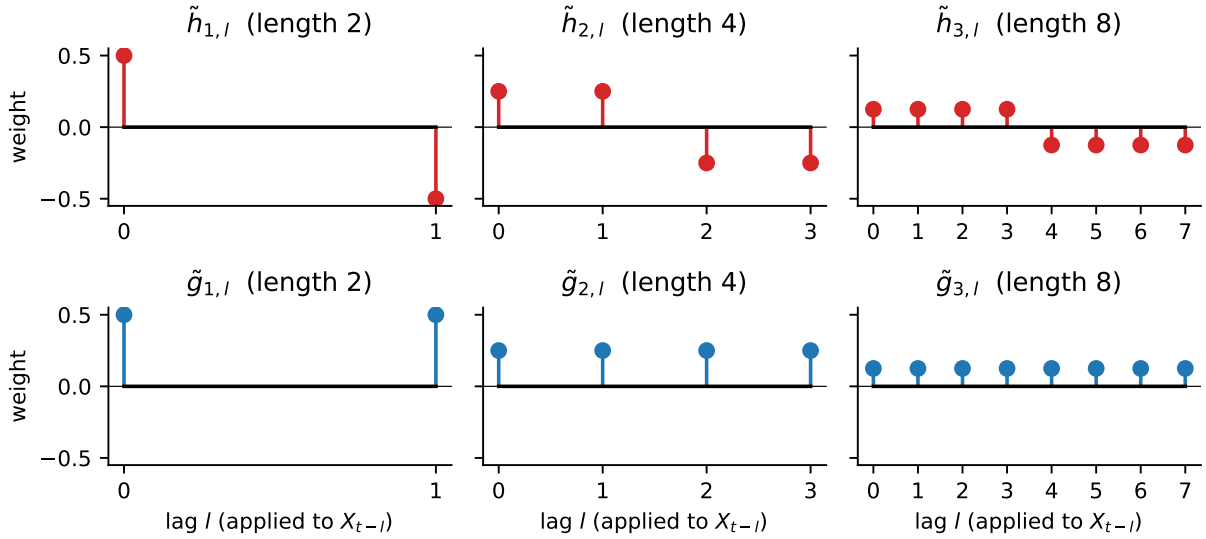


Figure 2: Equivalent Haar MODWT filters for levels 1–3. The wavelet filters (top) are "+block, −block" change detectors of growing width. The scaling filters (bottom) are plain moving averages of width 2^j.

5 Worked example: computing the MODWT by hand

Take $N = 8$ hourly $PM_{2.5}$ readings ($\mu g/m^3$) and decompose to $J = 3$ levels with the Haar MODWT:

$$X = (20,\ 22,\ 25,\ 24,\ 28,\ 30,\ 27,\ 26), \qquad t = 0, \ldots, 7.$$

Entries marked * use the circular wrap-around (e.g. $X_{-1} = X_7 = 26$).

5.1 Level 1: gap of 1 hour

Apply (5) with $j = 1$: compare each value with the previous one.

t	X_t	X_{t-1}	$W_{1,t} = \frac{X_t - X_{t-1}}{2}$	$V_{1,t} = \frac{X_t + X_{t-1}}{2}$	$W_{1,t} + V_{1,t}$
0	20	26*	−3	23	20
1	22	20	1	21	22
2	25	22	1.5	23.5	25
3	24	25	−0.5	24.5	24
4	28	24	2	26	28
5	30	28	1	29	30
6	27	30	−1.5	28.5	27
7	26	27	−0.5	26.5	26

Table 1: Level 1. Example: $W_{1,4} = (28 - 24)/2 = 2$ and $V_{1,4} = (28 + 24)/2 = 26$.

Notice $W_{1,t} + V_{1,t} = X_t$ for every t: the original value has been split exactly into a "change" part and an "average" part. The large value $W_{1,0} = -3$ is an artefact of the wrap-around: it compares $X_0 = 20$ with $X_7 = 26$, which are not really neighbours.

5.2 Level 2: gap of 2 hours, applied to V_1

Now we decompose the smooth series V_1, comparing values **two** steps apart.

t	$V_{1,t}$	$V_{1,t-2}$	$W_{2,t} = \frac{V_{1,t} - V_{1,t-2}}{2}$	$V_{2,t} = \frac{V_{1,t} + V_{1,t-2}}{2}$
0	23	28.5*	−2.75	25.75
1	21	26.5*	−2.75	23.75
2	23.5	23	0.25	23.25
3	24.5	21	1.75	22.75
4	26	23.5	1.25	24.75
5	29	24.5	2.25	26.75
6	28.5	26	1.25	27.25
7	26.5	29	−1.25	27.75

Table 2: Level 2. Example: $W_{2,5} = (29 - 24.5)/2 = 2.25$ and $V_{2,5} = (29 + 24.5)/2 = 26.75$.

Check against (6): $V_{2,5}$ should be the mean of the last four observations, $(30+28+24+25)/4 = 26.75$ ✓. And (7): $W_{2,5} = \frac{1}{2}\left[\frac{30+28}{2} - \frac{24+25}{2}\right] = \frac{1}{2}[29 - 24.5] = 2.25$ ✓.

5.3 Level 3: gap of 4 hours, applied to V_2

t	$V_{2,t}$	$V_{2,t-4}$	$W_{3,t} = \frac{V_{2,t} - V_{2,t-4}}{2}$	$V_{3,t} = \frac{V_{2,t} + V_{2,t-4}}{2}$
0	25.75	24.75*	0.5	25.25
1	23.75	26.75*	−1.5	25.25
2	23.25	27.25*	−2	25.25
3	22.75	27.75*	−2.5	25.25
4	24.75	25.75	−0.5	25.25
5	26.75	23.75	1.5	25.25
6	27.25	23.25	2	25.25
7	27.75	22.75	2.5	25.25

Table 3: Level 3. Example: $W_{3,7} = (27.75 - 22.75)/2 = 2.5$.

V_3 is the mean of the last $2^3 = 8$ values. Since $N = 8$ and the series is treated as circular, every window contains all eight observations, so $V_{3,t} = 202/8 = 25.25$ for all t: the coarsest smooth has collapsed to the overall mean. In real applications $N \gg 2^J$ and V_J is a genuine slowly varying trend.

5.4 All coefficients together

t	X_t	$W_{1,t}$	$W_{2,t}$	$W_{3,t}$	$V_{3,t}$	$W_1{+}W_2{+}W_3{+}V_3$
0	20	-3	-2.75	0.5	25.25	20
1	22	1	-2.75	-1.5	25.25	22
2	25	1.5	0.25	-2	25.25	25
3	24	-0.5	1.75	-2.5	25.25	24
4	28	2	1.25	-0.5	25.25	28
5	30	1	2.25	1.5	25.25	30
6	27	-1.5	1.25	2	25.25	27
7	26	-0.5	-1.25	2.5	25.25	26

Table 4: The complete Haar MODWT of X. Every row sums back to X_t.

Reading Table 4:

- W_1 is small and erratic: hour-to-hour jitter.
- W_2 goes from negative ($t = 0, 1$) to positive ($t = 3$–6): the series was rising over 2-hour blocks in the middle of the record.
- W_3 rises steadily from -2.5 to $+2.5$: the 4-hour block averages were climbing, which is the slower rise towards the peak at $t = 5$.
- V_3 carries the overall level.

A special property of Haar

For the Haar wavelet, $\tilde{h} + \tilde{g} = (1, 0)$, which implies $V_{j-1,t} = W_{j,t} + V_{j,t}$ at every level, and therefore

$$X_t = W_{1,t} + W_{2,t} + \cdots + W_{J,t} + V_{J,t}.$$

The coefficients themselves add up to the signal. This is *not* true for longer wavelets (Daubechies, least-asymmetric, ...). For those you need the multiresolution analysis of Section 7.

6 Inverting the MODWT: perfect reconstruction

The inverse MODWT (IMODWT) runs the pyramid backwards, from level J down to level 1:

$$V_{j-1,t} = \sum_{l=0}^{L-1} \tilde{h}_l \, W_{j,(t+2^{j-1}l) \bmod N} + \sum_{l=0}^{L-1} \tilde{g}_l \, V_{j,(t+2^{j-1}l) \bmod N}, \qquad j = J, J-1, \ldots, 1, \tag{8}$$

ending with $V_0 = X$. Note the $+$ in the index: the inverse looks *forward* in time. For Haar,

$$V_{j-1,t} = \tfrac{1}{2}(W_{j,t} - W_{j,t+2^{j-1}}) + \tfrac{1}{2}(V_{j,t} + V_{j,t+2^{j-1}}).$$

Example: rebuilding X_4 from level 1

From the level-1 table: $W_{1,4} = 2$, $W_{1,5} = 1$, $V_{1,4} = 26$, $V_{1,5} = 29$. Then

$$X_4 = \tfrac{1}{2}(2-1) + \tfrac{1}{2}(26+29) = 0.5 + 27.5 = 28 \checkmark$$

The inverse is exact: no information is lost by the MODWT. In WaveCatBoost this is the final step. The forecasted coefficients are passed through the IMODWT to produce the forecast of the pollutant:

$$\hat{X}_{N+h} = \text{IMODWT}(\hat{W}_{1,N+h}, \ldots, \hat{W}_{J,N+h}, \hat{V}_{J,N+h}).$$

7 Multiresolution analysis (MRA): details D_j and smooth S_J

The wavelet coefficients W_j are *measurements of change*. For interpretation and for an additive decomposition that works with *any* wavelet, we convert them back into the units of the original series. The **detail** at level j is what you get by inverting only W_j (setting all other coefficients to zero), and the **smooth** is what you get by inverting only V_J:

$$D_j = \text{IMODWT}(0, \ldots, 0, \underbrace{W_j}_{\text{level } j}, 0, \ldots, 0;\ 0), \qquad S_J = \text{IMODWT}(0, \ldots, 0;\ V_J).$$

Because the inverse is linear, these always add up exactly:

$$\boxed{X_t = \sum_{j=1}^{J} D_{j,t} + S_{J,t}} \tag{9}$$

This is the decomposition written in the WaveCatBoost paper: $Y_t = \sum_{k=1}^{K} D_{k,t} + S_{K,t}$.

For Haar, the first detail is $D_{1,t} = \frac{1}{2}(W_{1,t} - W_{1,t+1})$. It averages the change "into" t with the change "out of" t, which centres the detail on time t (a *zero-phase* filter).

t	X_t	$D_{1,t}$	$D_{2,t}$	$D_{3,t}$	$S_{3,t}$	$D_1{+}D_2{+}D_3{+}S_3$
0	20	−2	−1.875	−1.375	25.25	20
1	22	−0.25	−1.375	−1.625	25.25	22
2	25	1	−0.375	−0.875	25.25	25
3	24	−1.25	−0.125	0.125	25.25	24
4	28	0.5	0.875	1.375	25.25	28
5	30	1.25	1.875	1.625	25.25	30
6	27	−0.5	1.375	0.875	25.25	27
7	26	1.25	−0.375	−0.125	25.25	26

Table 5: Haar MRA of the example series. Example: $D_{1,0} = \frac{1}{2}(W_{1,0} - W_{1,1}) = \frac{1}{2}(-3-1) = -2$.

7.1 Coefficients W_j versus details D_j

	Wavelet coefficients $W_{j,t}$	**MRA details** $D_{j,t}$
What they measure	Change between block averages	The part of X_t at scale j, in μg/m^3
Time alignment	Lagged (filter looks backwards)	Centred on t (zero phase)
Uses data from	$X_t, X_{t-1}, \ldots$ (past only)	Past *and* future values
Adds up to X?	Only for Haar	Always
Energy preserved?	Yes (Section 8.1)	Not in general

The third row ("uses data from") matters a great deal for forecasting. We return to it in Section 12.2.

8 Key properties of the MODWT

8.1 Energy (variance) preservation

The MODWT preserves energy:

$$\|X\|^2 = \sum_{j=1}^{J} \|W_j\|^2 + \|V_J\|^2.$$

So the total variability of the series can be split into contributions from each scale. This is called the **wavelet variance**. For the example:

Quantity	Value
$\|X\|^2$	5174
$\|W_1\|^2$	20
$\|W_2\|^2$	28
$\|W_3\|^2$	25.5
$\|V_3\|^2$	5100.5
$\sum_j \|W_j\|^2 + \|V_3\|^2$	5174

Table 6: Energy decomposition of the example. Almost all energy is in V_3 because the series has a large mean (25.25). The fluctuating part has energy $20 + 28 + 25.5 = 73.5$, spread fairly evenly over the three scales.

The wavelet variance tells you *which time scales dominate* a pollutant's variability, which in turn suggests how many levels to use and which components are hardest to forecast.

8.2 Shift invariance

If the input is shifted in time, the MODWT coefficients shift by the same amount and are otherwise unchanged. The DWT does not have this property. Shift the example series circularly by one hour, $X' = (26, 20, 22, 25, 24, 28, 30, 27)$, and compare:

DWT		
k	$W_{1,k}$ of X	$W_{1,k}$ of X'
0	1.4142	−4.2426
1	−0.7071	2.1213
2	1.4142	2.8284
3	−0.7071	−2.1213

MODWT		
t	$W_{1,t}$ of X	$W_{1,t}$ of X'
0	−3	−0.5
1	1	−3
2	1.5	1
3	−0.5	1.5
4	2	−0.5
5	1	2
6	−1.5	1
7	−0.5	−1.5

Table 7: X' is X shifted by one hour. Left: DWT level-1 coefficients change completely when the input is shifted by one hour. Right: MODWT coefficients of the shifted series are simply the original coefficients moved down by one row.

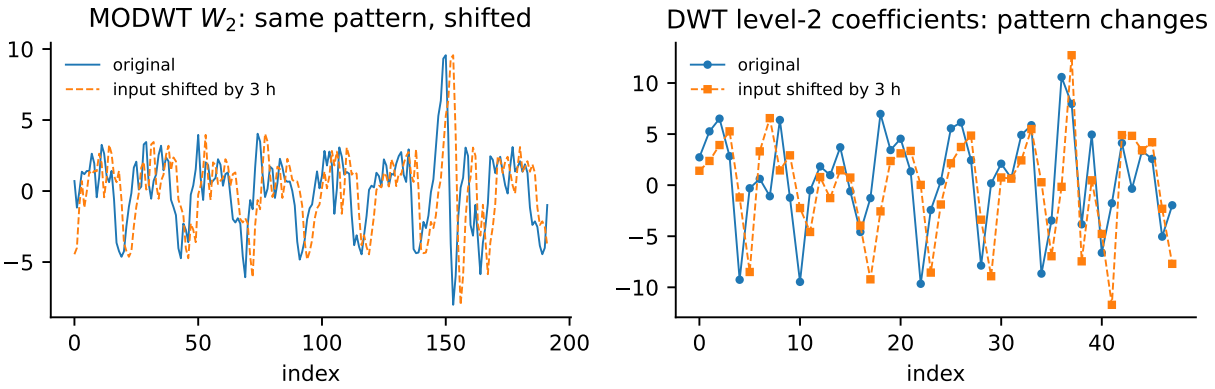


Figure 3: The same effect on a longer synthetic series shifted by 3 hours. The MODWT W_2 curves are identical apart from the shift. The DWT coefficients form a different pattern.

For forecasting, shift invariance means a pattern is described the same way no matter which hour it happens to start at. A model trained on these features therefore sees consistent inputs.

8.3 Works for any sample size

The MODWT is defined for every N, not just multiples of 2^J. The CPCB and ID1 series can be used as they are, without trimming or padding.

8.4 Redundancy

A level-J MODWT produces $(J+1)N$ numbers from N observations. This redundancy is what buys shift invariance and time alignment. The cost is $O(NJ)$ operations, which is negligible for hourly data.

8.5 Boundary coefficients

Because of the circular treatment, the first $L_j - 1$ coefficients of level j mix the beginning of the series with its end. For Haar, $L_j - 1 = 2^j - 1$: one coefficient at level 1, three at level 2, seven at level 3, and so on. These are the starred entries in the worked example. With a year of hourly data ($N \approx 8760$) and $J = 6$, only the first 63 values are affected. They are usually discarded before training.

9 Scale and frequency: what each level captures

Each MODWT wavelet filter is a **band-pass** filter. Level j is associated with the frequency band

$$\frac{1}{2^{j+1}} \le f \le \frac{1}{2^j} \quad \text{cycles per time step} \qquad \Longleftrightarrow \qquad \text{periods between } 2^j \text{ and } 2^{j+1} \text{ time steps.}$$

The smooth V_J (or S_J) holds everything with periods longer than 2^{J+1} steps.

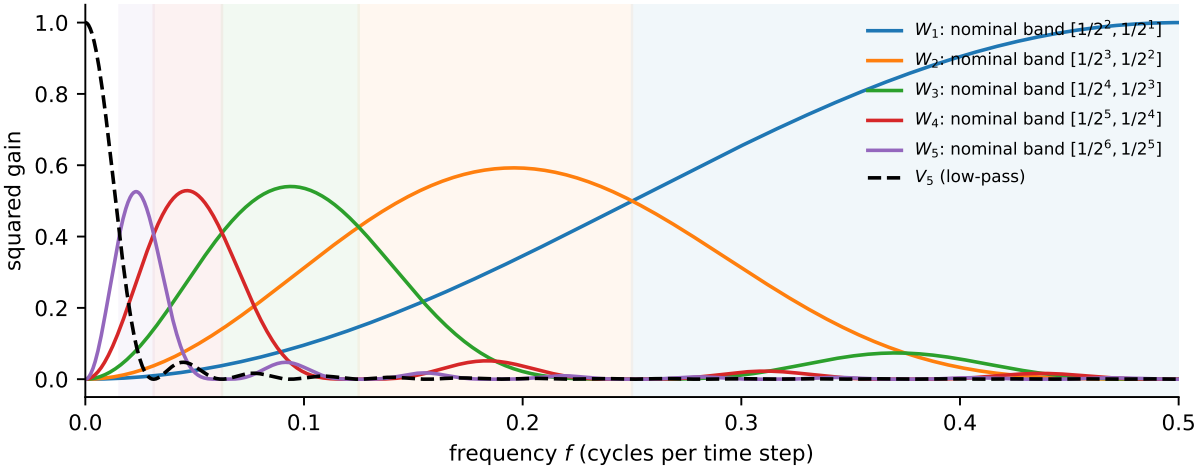


Figure 4: Squared gain of the Haar MODWT filters for levels 1–5, with the nominal bands shaded. Each level passes roughly one octave of frequencies. Haar filters are short, so their bands overlap considerably. Longer wavelets give sharper bands.

For **hourly** air-quality data this gives the following map:

Level	Physical scale 2^{j-1}	Periods captured	Typical air-quality content
D_1	1 h	2–4 h	sensor noise, short spikes
D_2	2 h	4–8 h	rush-hour peaks, short events
D_3	4 h	8–16 h	12-hour (semi-diurnal) cycle
D_4	8 h	16–32 h	**24-hour (diurnal) cycle**
D_5	16 h	32–64 h	2–3 day weather episodes
D_6	32 h	64–128 h	synoptic weather systems
S_6	–	$>$ 128 h	weekly cycle (168 h), seasonal trend

Table 8: Interpretation of the MODWT levels for hourly data.

9.1 Choosing the number of levels J

- **Upper bound**: 2^J must be well below N, typically $J \le \lfloor \log_2 N \rfloor$ and in practice much smaller.
- **Physical reasoning**: choose J so that the important cycles are separated. For hourly data, $J \ge 4$ isolates the daily cycle in D_4.
- **Boundary cost**: each extra level doubles the number of boundary-affected points.
- **Model cost**: in WaveCatBoost, each level adds one CatBoost model.

10 A realistic example: synthetic hourly $PM_{2.5}$

To see the MODWT at work on something that looks like real data, we build a 32-day hourly series:

$$X_t = \underbrace{45 + 0.012t}_{\text{trend}} + \underbrace{9\sin\tfrac{2\pi(t-6)}{24} + 3.5\sin\tfrac{2\pi t}{12}}_{\text{daily and half-daily cycles}} + \underbrace{5\sin\tfrac{2\pi t}{168}}_{\text{weekly}} + \underbrace{\varepsilon_t}_{\text{noise, } \sigma=2.5} + \underbrace{3 \text{ short spikes}}_{\text{events}}.$$

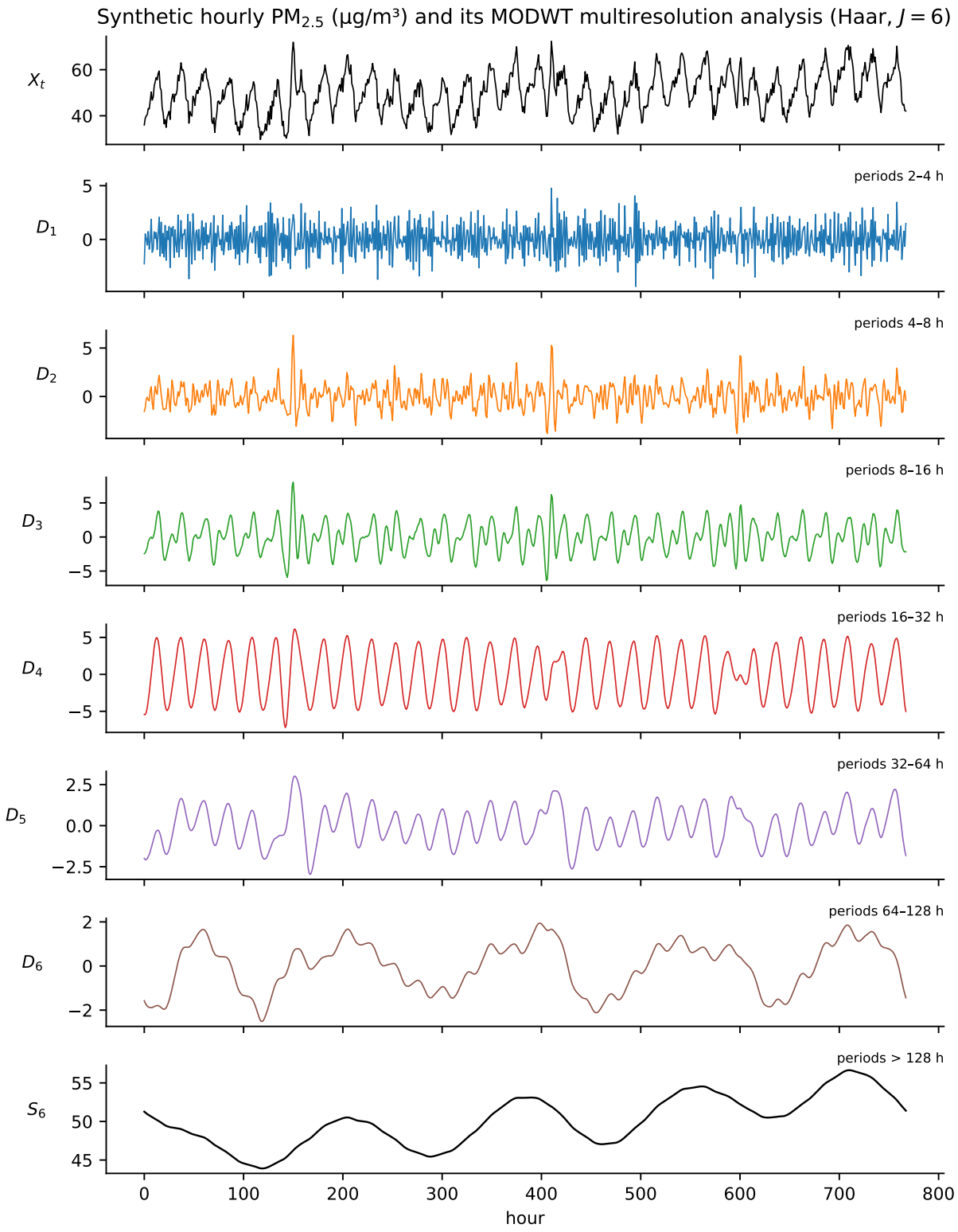


Figure 5: Haar MODWT multiresolution analysis of the synthetic series ($J = 6$). The components add up exactly to the top panel.

Reading Figure 5 against Table 8:

- D_1, D_2 contain the noise and the sharp parts of the three spikes (hours 150, 410, 600), which show up as localised bursts. The Fourier transform could not localise them in this way.

- D_3 picks up the 12-hour cycle.
- D_4 is dominated by a clean oscillation with a 24-hour period: the daily cycle.
- D_5, D_6 contain leakage from neighbouring bands (Haar's bands overlap) and slow wiggles.
- S_6 holds the weekly cycle riding on the upward trend.

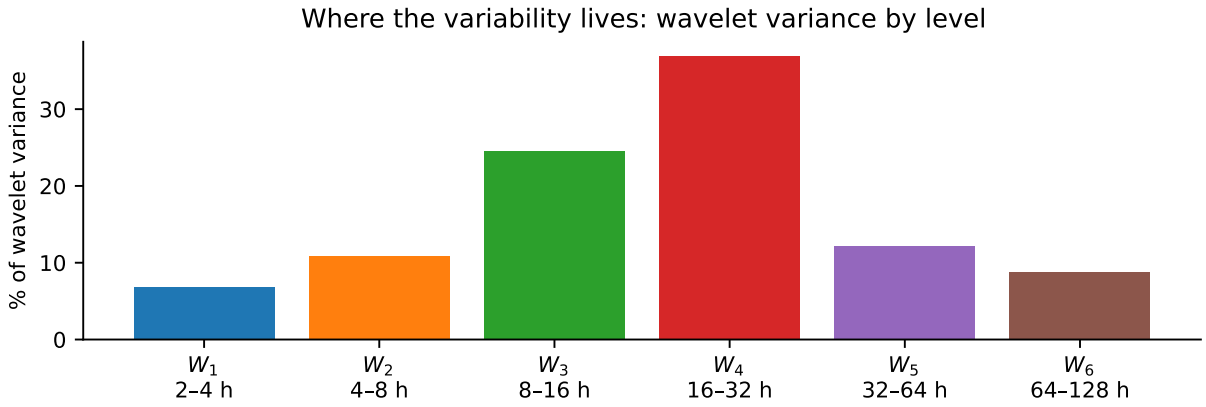


Figure 6: Wavelet variance by level for the synthetic series. Level 4 dominates because the 24-hour cycle has the largest amplitude.

Each component is far simpler than X_t: D_4 is almost a sine wave, S_6 is smooth and slowly varying, and D_1 is mostly noise. This is exactly why forecasting each component separately can beat forecasting X_t directly.

11 Choosing the wavelet

Wavelet	Advantages	Disadvantages
Haar ($L = 2$)	Simplest; shortest filter, so fewest boundary points; coefficients use only very recent data; coefficients add up to X	Blocky; frequency bands overlap a lot
Daubechies D(4), D(6)	Smoother; better frequency separation	Longer filters, so more boundary points
Least asymmetric LA(8)	Near-linear phase, so features align well in time; common choice in time-series work	Filter length 8, so $L_j - 1 = 7(2^j - 1)$ boundary points

WaveCatBoost uses Haar. The paper does not report an ablation over wavelet families, so trying D(4) or LA(8) would be a natural experiment in a replication.

12 MODWT inside WaveCatBoost

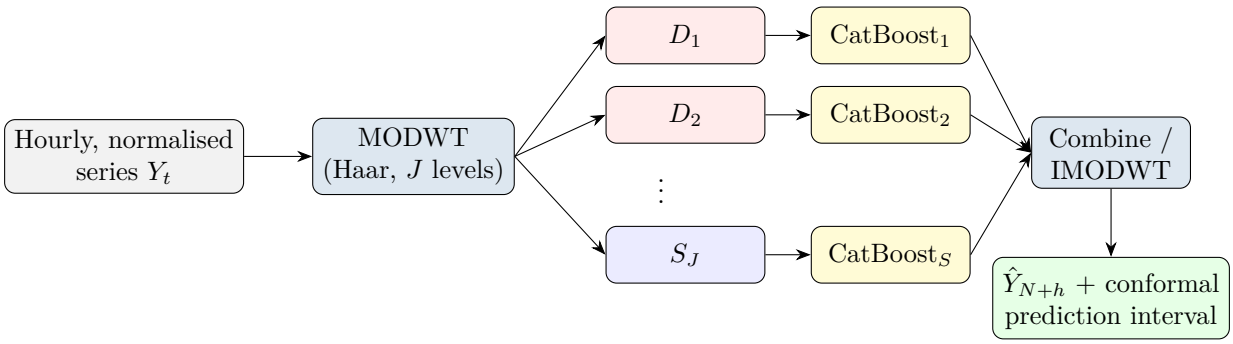


Figure 7: Role of the MODWT in WaveCatBoost.

12.1 Step by step

1. **Decompose** the normalised series Y_t into $D_{1,t}, \ldots, D_{J,t}, S_{J,t}$.
2. **Build lag features** for each component, which works because the MODWT keeps every component on the hourly time axis:

$$\hat{D}_{j,t} = f_j(D_{j,t-1}, D_{j,t-2}, \ldots, D_{j,t-p}), \qquad \hat{S}_{J,t} = f_S(S_{J,t-1}, \ldots, S_{J,t-p}),$$

   where each f_j is a separate CatBoost regressor.
3. **Forecast** each component recursively for $h = 24, 168, 336, 744$ steps (1, 7, 14, 31 days).
4. **Reconstruct**: add the component forecasts (Haar / MRA) or apply the IMODWT (coefficients).
5. **Quantify uncertainty** around $\hat{Y}$ with conformal prediction.

Toy forecast

Suppose that, for the next hour, the component models predict $\hat{D}_1 = 0.8$, $\hat{D}_2 = 1.4$, $\hat{D}_3 = 1.1$ and $\hat{S}_3 = 26.0$. With an additive (MRA) decomposition the forecast is simply $\hat{X} = 0.8 + 1.4 + 1.1 + 26.0 = 29.3$ μg/m^3. Each model had an easier job: $\hat{S}_3$ only needs to follow a smooth level, and $\hat{D}_1$ only needs to predict small hour-to-hour jitter. (These numbers are illustrative only and are not taken from the paper.)

12.2 A critical pitfall: look-ahead leakage

Decomposing the full series before splitting causes leakage

MRA details $D_{j,t}$ are computed with the inverse transform (8), which looks **forward** in time. If you decompose the entire dataset first and then split it into training and test sets, the training features at time t already contain information from $t+1, t+2, \ldots$, including test-period values. Reported accuracy can then be far better than anything achievable in real-time forecasting.

The example below makes this concrete. We compute the level-j value at $t = 6$ twice: once using the full series $X_0, \ldots, X_7$, and once using only the data available at that moment, $X_0, \ldots, X_6$.

Value at $t = 6$	using $X_0, \ldots, X_7$ (full)	using $X_0, \ldots, X_6$ only
$D_{1,6}$	-0.5	1
$D_{2,6}$	1.375	0.5625
$D_{3,6}$	0.875	0.2656
$W_{1,6}$	-1.5	-1.5
$W_{2,6}$	1.25	1.25
$W_{3,6}$	2	1.875

Table 9: The MRA details at $t = 6$ change when the future value X_7 is removed, so they depend on the future. The Haar coefficients $W_{1,6}$ and $W_{2,6}$ do not change, because they only use past data. $W_{3,6}$ changes only because level 3 needs 8 past values and just 7 exist, which triggers the circular boundary.

Safe practice for forecasting experiments:

- Split the data chronologically *first*, then decompose only the training portion.
- For the test period, use a **rolling (expanding-window) decomposition**: at each forecast origin, decompose only the data observed so far and take the last values as features.
- Alternatively, use the Haar *coefficients* $W_{j,t}$ and $V_{J,t}$, which are causal by (6)–(7), and reconstruct with the IMODWT. Discard the first $2^J - 1$ boundary-affected points.

13 Implementation in Python

13.1 From scratch (Haar)

The Haar MODWT needs only `numpy.roll`. The code below was used to produce the tables in this document.

```python
import numpy as np

def haar_modwt(x, J):
    """Circular Haar MODWT. Returns [W_1, ..., W_J] and V_J."""
    V = np.asarray(x, dtype=float)
    W = []
    for j in range(1, J + 1):
        s = 2 ** (j - 1)
        V_lag = np.roll(V, s) # V_{j-1, t-s}
        W.append((V - V_lag) / 2) # wavelet (detail) coefficients
        V = (V + V_lag) / 2 # scaling (smooth) coefficients
    return W, V

def haar_inverse_step(W_j, V_j, j):
    """Recover V_{j-1} from W_j and V_j."""
    s = 2 ** (j - 1)
    return (W_j - np.roll(W_j, -s) + V_j + np.roll(V_j, -s)) / 2

def haar_imodwt(W, V_J):
    V = V_J
    for j in range(len(W), 0, -1):
        V = haar_inverse_step(W[j - 1], V, j)
    return V

def haar_mra(x, J):
    """Details D_1..D_J and smooth S_J with x = sum(D) + S."""
```

```
    W, V_J = haar_modwt(x, J)
    zeros = [np.zeros_like(w) for w in W]
    D = [haar_imodwt(zeros[:j] + [W[j]] + zeros[j + 1:], np.zeros_like(V_J))
         for j in range(J)]
    S = haar_imodwt(zeros, V_J)
    return D, S

x = np.array([20, 22, 25, 24, 28, 30, 27, 26], dtype=float)
W, V3 = haar_modwt(x, 3)
print(W[0]) # [-3. 1. 1.5 -0.5 2. 1. -1.5 -0.5]
print(V3) # [25.25 25.25 ... 25.25]
```

13.2 With PyWavelets

PyWavelets implements the same decomposition through its stationary wavelet transform. We checked that the following gives results identical to `haar_mra` above:

```
import pywt
components = pywt.mra(x, "haar", level=3, transform="swt")
S3, D3, D2, D1 = components # ordered from coarsest to finest
```

In R, the `waveslim` package (`modwt`, `mra`) and the `wavelets` package (`modwt`, `mra`) implement the transform as defined by Percival and Walden.

14 Summary

MODWT cheat sheet

- **What**: a wavelet transform that splits a series into J detail components and one smooth component, each the same length as the original.
- **How**: repeatedly filter the smooth series with a high-pass $\tilde{h}$ and low-pass $\tilde{g}$ filter, doubling the gap between filter taps at each level. There is no downsampling.
- **Haar version**: $W_{j,t} = \frac{1}{2}(V_{j-1,t} - V_{j-1,t-2^{j-1}})$, $V_{j,t} = \frac{1}{2}(V_{j-1,t} + V_{j-1,t-2^{j-1}})$.
- **Additive**: $X_t = \sum_j D_{j,t} + S_{J,t}$ exactly; for Haar, also $X_t = \sum_j W_{j,t} + V_{J,t}$.
- **Energy**: $\|X\|^2 = \sum_j \|W_j\|^2 + \|V_J\|^2$, which gives a variance-by-scale breakdown.
- **Scales**: level j holds periods of 2^j to 2^{j+1} steps; for hourly data, D_4 holds the daily cycle.
- **Advantages over the DWT**: any N, shift invariant, time-aligned components, suited to lag features.
- **Watch out for**: circular boundary effects ($2^j - 1$ points per level for Haar) and look-ahead leakage when MRA details are computed on the full dataset.

References

[1] D. B. Percival and A. T. Walden, *Wavelet Methods for Time Series Analysis*. Cambridge University Press, 2000. (The standard reference for the MODWT; Chapter 5 covers the definitions used here.)

[2] S. Mallat, *A Wavelet Tour of Signal Processing*, 3rd ed. Academic Press, 2008.

[3] J. Borah, T. Chakraborty, M. S. M. Nadzir, M. G. Cayetano, F. Benedetto and S. Majumdar, "A Novel Hybrid Approach for Efficiently Forecasting Air Quality Data (WaveCatBoost)," *IEEE Sensors Letters*, 2025.

[4] G. R. Lee *et al.*, "PyWavelets: A Python package for wavelet analysis," *Journal of Open Source Software*, 4(36), 1237, 2019.